


# Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

*A right Choice for the Real Aspirant*

**ICON Central Office - Madhapur - Hyderabad**

**Sec:** Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -1(Adv-2019-P1-Model)** **Date: 13-05-2026**

**Time: 09.00Am to 12.00Pm** **GTA-32** **Max. Marks: 186**

13-05-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2019-P1)_GTA-32_Syllabus,

**PHYSICS : TOTAL SYLLABUS**

**CHEMISTRY : TOTAL SYLLABUS**

**MATHEMATICS : TOTAL SYLLABUS**

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**Name of the Student:** _____ **H.T. NO:** | | | | | | | |



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# JEE-ADVANCED-2019-P1-Model

Time: 3:00Hr's **IMPORTANT INSTRUCTIONS** Max Marks: 186

## PHYSICS:

| Section | Question Type | +Ve Marks | - Ve Marks | No.of Qs | Total marks |
|---|---|---|---|---|---|
| Sec – I(Q.N : 1 – 4) | Questions with Single Correct Choice | 3 | -1 | 4 | 12 |
| Sec – II(Q.N : 5 – 12) | Questions with Multiple Correct Choice **+1 partial marks** | 4 | -1 | 8 | 32 |
| Sec – III(Q.N : 13 – 18) | Questions with Numerical Value Answer Type | 3 | 0 | 6 | 18 |
| **Total** | | | | **18** | **62** |

## CHEMISTRY:

| Section | Question Type | +Ve Marks | - Ve Marks | No.of Qs | Total marks |
|---|---|---|---|---|---|
| Sec – I(Q.N : 19 – 22) | Questions with Single Correct Choice | 3 | -1 | 4 | 12 |
| Sec – II(Q.N : 23 – 30) | Questions with Multiple Correct Choice **+1 partial marks** | 4 | -1 | 8 | 32 |
| Sec – III(Q.N : 31 – 36) | Questions with Numerical Value Answer Type | 3 | 0 | 6 | 18 |
| **Total** | | | | **18** | **62** |

## MATHEMATICS:

| Section | Question Type | +Ve Marks | - Ve Marks | No.of Qs | Total marks |
|---|---|---|---|---|---|
| Sec – I(Q.N : 37 – 40) | Questions with Single Correct Choice | 3 | -1 | 4 | 12 |
| Sec – II(Q.N : 41 – 48) | Questions with Multiple Correct Choice **+1 partial marks** | 4 | -1 | 8 | 32 |
| Sec – III(Q.N : 49 – 54) | Questions with Numerical Value Answer Type | 3 | 0 | 6 | 18 |
| **Total** | | | | **18** | **62** |

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**PHYSICS** **Max Marks: 62**

## SECTION – I
## (SINGLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct. **Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases. Section 1 (Max Marks: 12)**

- Section 1 contains Four questions
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If the answer is correct)
  Zero Marks: **0** (If the question is unanswered)
  Negative Marks: -1 (In all other cases)

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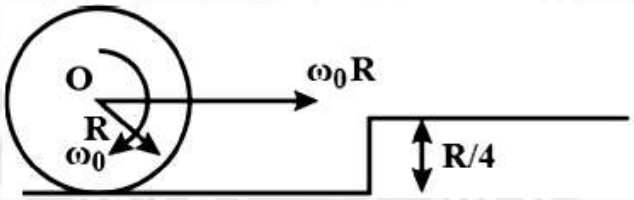
**1.** A particle is moving along path $y = x^2$ in X – Y plane with constant speed of 1 m/s. What will be its magnitude of acceleration when it is at $x = 0$. (Assume all the units are in SI)

**A)** Zero **B)** $2\ m/s^2$ **C)** $20\ m/s^2$ **D)** $200\ m/s^2$

**2.** The power output from a certain experimental car which is designed in shape like a cube is proportional to the mass $m$ of the car. The force of air friction on the car is proportional to $Av^2$, where $v$ is the speed of the car and A the cross sectional area. On a level surface the car has a maximum speed $v_{max}$. Assuming that all geometrically similar versions of this design have the same density, then which of the following is **TRUE**?

**A)** $v_{max} \propto m^{1/9}$ **B)** $v_{max} \propto m^{1/7}$ **C)** $v_{max} \propto m^{1/3}$ **D)** $v_{max} \propto m^{2/9}$

**3.** A uniform solid cylinder of mass $m$ and radius $R$ is rolling without slipping on a horizontal surface with angular velocity $\omega_0$. The velocity of centre of mass of cylinder is $\omega_0 R$. The cylinder comes across a step of height $R/4$. (Assume required friction is present at edge of the step)




Find the maximum angular velocity $(\omega_0)$ of cylinder so that cylinder rides up the step without slipping or jumping.

**A)** $\frac{3}{5}\sqrt{\frac{3g}{R}}$ **B)** $\frac{2}{5}\sqrt{\frac{3g}{R}}$ **C)** $\frac{1}{2}\sqrt{\frac{3g}{R}}$ **D)** $\frac{5}{3}\sqrt{\frac{3g}{R}}$



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**4.** A smooth narrow tunnel is dug inside the earth along a diameter PQ. A particle of mass '$m$' is placed at centre of tunnel and another particle of mass '$m$' is projected with a velocity $v_0$ into the tunnel from $P$ (surface). Mass and radius of earth are $M$ and $R$ respectively. The collision between the particles is perfectly inelastic. Assume $M >> m$ and neglect gravitational interaction between the particles. What is the minimum value of $v_0$ so that the combined mass $2m$ can escape from earth's gravity?

**A)** $\sqrt{\frac{5GM}{R}}$ **B)** $\sqrt{\frac{5GM}{2R}}$ **C)** $\sqrt{\frac{11GM}{R}}$ **D)** $\sqrt{\frac{11GM}{2R}}$

## SECTION – II
## (ONE OR MORE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. **Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -1 in all wrong cases**

**Section 2 (Max Marks: 32)**

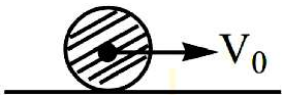
- Section two contains Eight (8) questions
- Each Question has Four Options and One Or More Than One of these four will be the correct answer.
- For each question, choose the option(s) corresponding to all the correct answer(s)

The Marking scheme to evaluate Answer to each question will be :
Full Marks: **+4** (If the answer is correct)
Partial Marks: **+3** (If all the four options are correct but **only 3** are chosen)
Partial Marks: **+2** (If THREE or More options are correct but **only 2** correct answers are chosen)
Partial Marks: **+1** (If TWO or More options are correct but **only 1** correct answer is chosen)
Zero Marks: **0** (If the question is unanswered)
Negative Marks: -1 (In all other cases)

**5.** Two towns A and B are connected by a regular bus service with a bus leaving in either direction every T minutes. A man cycling with a speed 20 km/hr in the direction A to B notices that a bus goes past him every 18 minutes in the direction of his motion and every 6 min in the opposite direction. Assume the buses ply with constant speed V on the road.

**A)** T = 6 min **B)** T = 9 min **C)** V = 40 km/hr **D)** V = 60 km/hr

**6.** A uniform disc of mass $m$ and radius $R$ starts with velocity $V_0$ on rough horizontal floor with a pure translational motion at $t = 0$ as shown. At $t = t_0$, disc starts rolling without sliding. Then (Coefficient of friction is $\mu$)






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**A)** Net work done by frictional force upto time $t \le t_0$ is given by $\frac{m\mu gt}{2}(3\mu gt - 2V_0)$

**B)** Net work done by frictional force upto time $t \le t_0$ is given by $\frac{m\mu gt}{2}(2\mu gt - V_0)$

**C)** Net work done by frictional force upto time $t = 2t_0$ is given by $m\mu gt_0(2\mu gt_0 - V_0)$

**D)** Net work done by frictional force upto time $t = 2t_0$ is given by $\frac{m\mu gt_0}{2}(3\mu gt - 2V_0)$

**7.** Two large parallel metal plates are located in vacuum. One of these serve as cathode, a source of electron with negligible initial speed. An electron flow directed toward the opposite plate produces a space charge causing the potential in the gap between the plate vary as $V = ax^{4/3}$, where a is a +ve constant, and $x$ is distance from cathode. Then

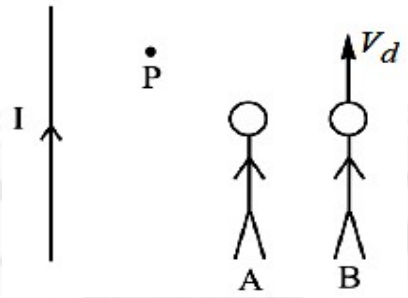
**A)** Magnitude of electric field is proportional to $x^{1/3}$

**B)** Volume charge density is proportional to $x^{-2/3}$

**C)** Current density is proportional to $x^{1/3}$

**D)** Current density is independent of $x$

**8.** There is long straight conducting wire in which a steady electric current I is flowing. There are two observers A and B observing electric field $\overline{E}$ and magnetic field $\overline{B}$ at a point P near the conducting wire. Observer A is stationary whereas observer B is moving with drift speed of electron in the wire in the direction of electric current in it. Which options are **CORRECT**?




**A)** Electric field observed by A and B are same.

**B)** Electric field observed by A is zero, whereas by B is non–zero.

**C)** Magnetic field observed by A and B are same.

**D)** Magnetic field observed by B is zero, whereas by A is non–zero.



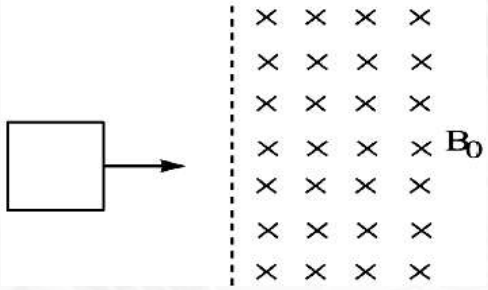
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**9.** A current $I$ flows in a long single layer solenoid with cross–sectional radius $R$. The number of turns per unit length of the solenoid equals $n$. The current at which the winding may rupture if the tensile strength of the wire is equal to $F_\ell$.

**A)** $\frac{1}{2}\sqrt{\frac{F_\ell}{\mu_0 nR}}$ **B)** $\sqrt{\frac{F_\ell}{\mu_0 nR}}$ **C)** $2\sqrt{\frac{F_\ell}{\mu_0 nR}}$ **D)** $4\sqrt{\frac{F_\ell}{\mu_0 nR}}$

**10.** A square loop of side a, resistance $R$, mass $m$ is sliding as shown on a smooth horizontal table with speed $v_0$. It enters a uniform magnetic field of magnitude $B_0$ perpendicular to the table. It is seen that the loop comes to rest after entering a distance $\ell$ inside the magnetic field. The value of $v_0$ can be




**A)** $\frac{2B^2a^3}{mR}$ **B)** $\frac{B^2a^3}{2mR}$ **C)** $\frac{B^2a^3}{3mR}$ **D)** $\frac{B^2a^3}{6mR}$

**11.** If 1 kg of water freezes, then 334 kJ of energy are released. The energy released if 1 kg of super cooled water at $-10^\circ C$ freezes is Q. Assume that during the freezing process the temperature remains constant, and the specific heat of water is $4.2\,kJ\,kg^{-1}K^{-1}$ and constant and specific heat of ice is $2.1\,kJ\,kg^{-1}K^{-1}$ which is also constant.

**A)** The value of Q is 313 kJ

**B)** The value of Q is 334 kJ

**C)** The change in internal energy of ice obtained by this process is not equal to the change internal energy of ice obtained by freezing water at $0^\circ C$ and then cooled upto $-10^\circ C$.

**D)** The change in internal energy of ice obtained by this process is equal to the change in internal energy of ice obtained by freezing water at $0^\circ C$ and then cooled upto $-10^\circ C$.



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**12.** The particle displacement of a travelling longitudinal wave is represented by $S = S(x,t)$. The midpoints of a compression zone and an adjacent rarefaction zone are represented by the letter 'C' and 'R'. Which of the following is/are **TRUE**?

**A)** $|\partial S / \partial x|_C = |\partial S / \partial x|_R$

**B)** $|\partial S / \partial t|_C = |\partial S / \partial t|_R = 0$

**C)** $(pressure)_C - (pressure)_R = 2|\partial S / \partial x|_C \times$ Bulk modulus of air

**D)** Particles of air are stationary at mid–way between 'C' and 'R'.

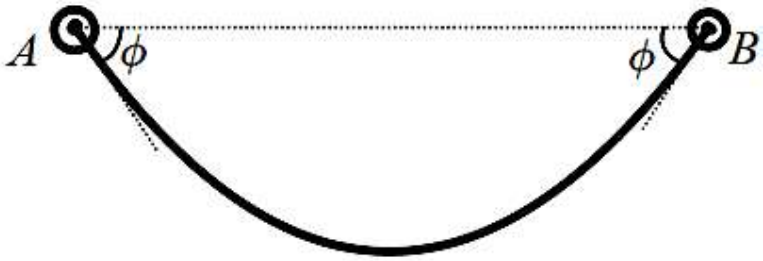
## SECTION – III
## (NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. **(e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).**
**Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 3 (Max Marks: 18)**

- This Section contains **Six (06)** Questions. The answer to each question is a Numerical Value
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If ONLY the correct numerical value is entered)
  Zero Marks: 0 (In all other cases)

**13.** The nuclear reaction James Chadwick used in his experiment to identify the neutron was $\alpha + {}^{11}B \rightarrow n + {}^{14}N$. Assume that the $\alpha$ particle had energy 5.3 MeV and that the ${}^{11}B$ was at rest; $M\left({}^{11}B\right) = 11.009305u$. If the kinetic energy of the ${}^{14}N$ is 0.8 MeV, what kinetic energy would the neutron have (in MeV)? (Given masses are that of nucleus) $M_{He} = 4.00260u, M_n = 1.00867u, M_N = 14.00307u, 1\ amu = 931MeV$.

**14.** A uniform rope of length $\ell = 32m$ is suspended between two fixed nails A and B that are in the same horizontal level. If the rope makes $\phi = 53^\circ$ with the horizontal at the nails, find the radius of curvature (in m) of catenary formed by the rope at a point where tangent is inclined at $\theta = 30^\circ$ to the horizontal.



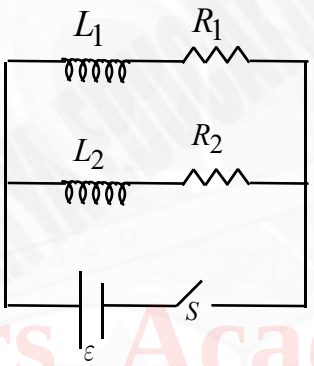



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**15.** A person is standing on a horizontal platform which is rotating with angular velocity $\omega = 1\ rad/s$ in anticlockwise direction about a fixed vertical axis. Distance of person from the axis of rotation is 4 m. He projects a ball with speed 7.5 m/s in vertically upward direction relative to himself. When the ball is at topmost point, what is acceleration (in $m/s^2$) of ball with respect to the man on platform $\left(g = 10\,m/s^2\right)$.

**16.** A uniform rod of mass m = 1 kg and length $\ell = 2m$ is rotating in horizontal plane on a smooth horizontal table with uniform angular velocity $\omega = 1\ rad/s$ about a vertical axis passing through one of it's ends. Find the horizontal force exerted by one half of the rod on another half (in N)

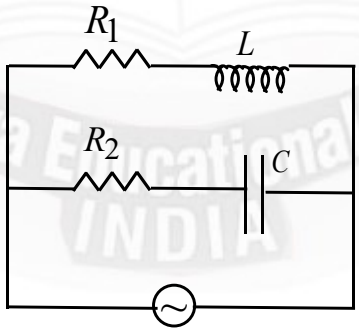
**17.** Switch S was kept closed for a long time in the circuit shown. At t = 0, switch was open. Find the current through the inductor $L_1$, immediately after the switch is open (in A)




$L_1 = L$
$L_2 = 2L$
$\varepsilon = 10V$
$R_1 = 3\Omega$
$R_2 = 5\Omega$

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**18.** Resistances $R_1$ and $R_2$ are $5\Omega$ and $5\sqrt{3}\,\Omega$ respectively. Reactance of inductor is $5\sqrt{3}\,\Omega$ and of capacitor is $5\Omega$. Source voltage oscillates as $V = 10\sqrt{2}\sin\omega t V$. Average power delivered by source is (in watt)





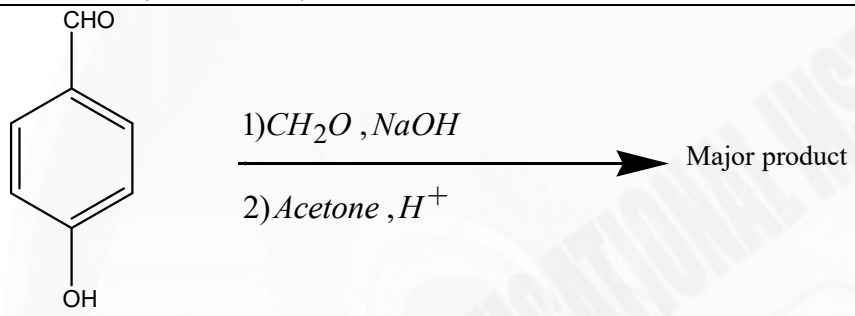

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**CHEMISTRY** **Max Marks: 62**

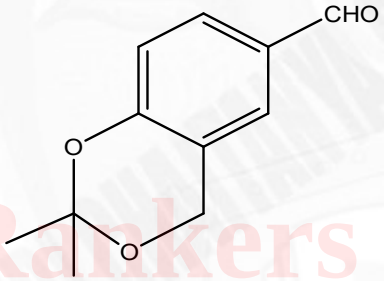
## SECTION – I
## (SINGLE CORRECT ANSWER TYPE)

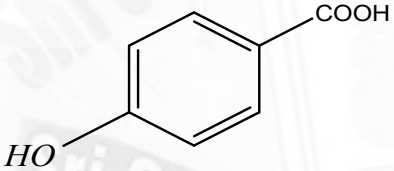
This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct. **Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases. Section 1 (Max Marks: 12)**

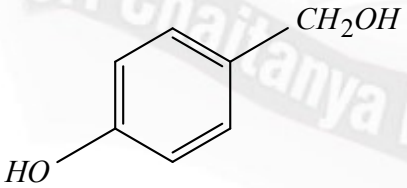
- Section 1 contains Four questions
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If the answer is correct)
  Zero Marks: **0** (If the question is unanswered)
  Negative Marks: -1 (In all other cases)

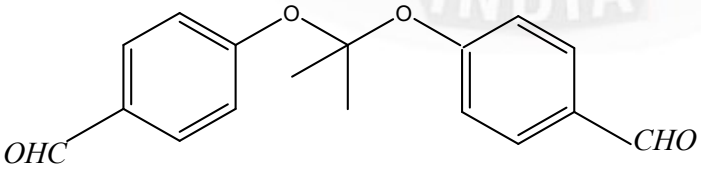



**19.** What is the major product of above reaction?

**A)** 


**B)** 


**C)** 


**D)** 




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**20.** A 0.14 mg sample of a long-chain fatty acid $\left(M = 280\,g\,mol^{-1}\right)$ is spread on a water surface to form a condensed monolayer. The molecules are packed such that their spherical polar heads form a two-dimensional hexagonal close-packed layer. If the distance of closest approach between the centers of two adjacent polar heads is $a = 5\,\mathring{A}$, what is the total surface area covered by the monolayer (in $cm^2$)?

Use: $N_a = 6\times10^{23}\,mol^{-1}$ $\sqrt{3} = 1.73$

**A)** $3.25\times10^2$ **B)** $4.87\times10^2$ **C)** $6.49\times10^2$ **D)** $8.12\times10^2$

**21.** A crystalline salt **(X)** is used in the laboratory to identify transition metal cations.When **(X)** is heated strongly, it loses weight and swells into a white porous mass **(Y)**, which further melts to form a transparent glassy bead **(Z)**. A piece of the glassy bead **(Z)** is dipped in a solution of a copper salt and heated in a reducing flame to give red and opaque. Which of the following statements is **incorrect** regarding the species involved?

**A)** Compound **(X)** contains two $sp^2$ and two $sp^3$ hybridized boron atoms in its anionic part.

**B)** The glassy bead **(Z)** is a mixture of sodium metaborate $\left(NaBO_2\right)$ and boric anhydride $\left(B_2O_3\right)$.

**C)** The red color in the reducing flame is due to the formation of metallic copper.

**D)** Treatment of an aqueous solution of **(X)** with a strong mineral acid like HCl results in the precipitation of weak protic acid.

**22.** Select the correct group of statements from the list related to the separation techniques.

**I)** Multiple small solvent extractions recover solute more efficiently than a single large extraction.

**II)** A fractionating column provides repeated vaporization–condensation, to separate liquids

**III)** Reduced pressure distillation lowers glycerol's boiling point, preventing its thermal decomposition into acrolein and water**.**

**IV)** In steam distillation of nitrophenol isomers, the internally hydrogen-bonded isomer remains in the residue.

**A)** II, III only **B)** II,III,IV only **C)** I, II, III only **D)** I, IV only



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## SECTION – II
## (ONE OR MORE CORRECT ANSWER TYPE)

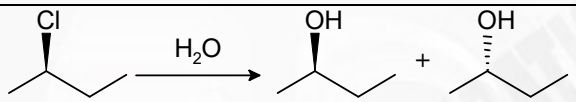
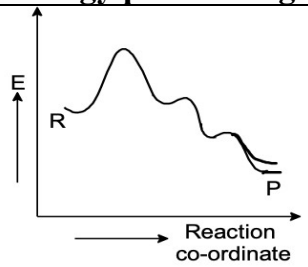
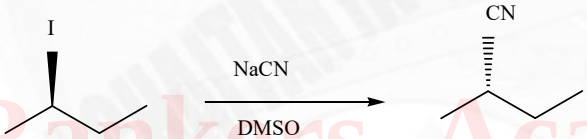
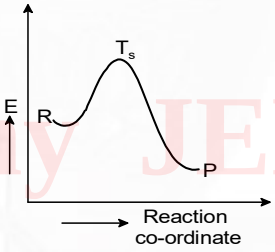
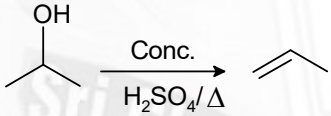
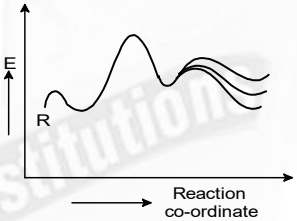
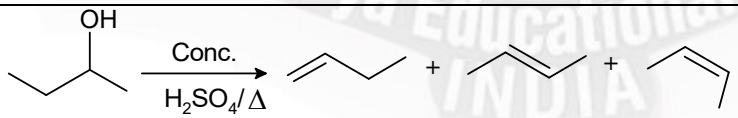
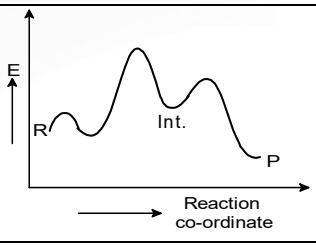
This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. **Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -1 in all wrong cases**

**Section 2 (Max Marks: 32)**

- Section two contains Eight (8) questions
- Each Question has Four Options and One Or More Than One of these four will be the correct answer.
- For each question, choose the option(s) corresponding to all the correct answer(s)

The Marking scheme to evaluate Answer to each question will be :
Full Marks: **+4** (If the answer is correct)
Partial Marks: **+3** (If all the four options are correct but **only 3** are chosen)
Partial Marks: **+2** (If THREE or More options are correct but **only 2** correct answers are chosen)
Partial Marks: **+1** (If TWO or More options are correct but **only 1** correct answer is chosen)
Zero Marks: **0** (If the question is unanswered)
Negative Marks: -1 (In all other cases)

**23.** Select the option in which the reaction is correctly matched with the qualitative energy profile diagram shown alongside it. (E = enthalpy).

| | Reaction | Energy profile diagram |
|---|---|---|
| **A)** |   |   |
| **B)** |   |   |
| **C)** |   |   |
| **D)** |   |   |



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**24.** Consider the oxyacids $HNO_2, H_2SO_4$ and $HC\ell O_3$. Which of the following statements is/are correct?

**A)** The geometry determining electron pairs of the central atom follows the order $HNO_2 < HC\ell O_3 = H_2SO_4$,

**B)** $HNO_2$ and $H_2SO_4$ can be generated by hydration of their respective acidic anhydrides.

**C)** The number of terminal oxo (M=O) linkages and number of $p\pi - d\pi$ bonds are identical in $HC\ell O_3$ and $H_2SO_4$.

**D)** Hydrogen gas is not produced in the reaction between copper and oxyacid.

**25.** Solid iodine is heated with an aqueous solution of sodium carbonate. The reaction produces two soluble ionic species **X** and **Y**. Species **X** forms a bright yellow precipitate with lead(II) acetate solution. Which of the following observations is/are consistent with the identity and chemical behaviour of **Y**?

**A)** On acidification in the presence of iodide ions, a brown solution develops due to liberation of elemental iodine.

**B)** In alkaline medium, **Y** oxidises dichromate ions to chromate ions.

**C)** On treatment with hydrogen sulfide in acidic solution, a yellow precipitate is formed.

**D)** In acidic medium, **Y** is reduced by sulfur dioxide

**26.** Consider a homonuclear diatomic molecule $X_2$ formed by second-period atoms. The internuclear axis is taken as the z-axis. Molecular orbitals are formed by linear combination of atomic orbitals in the usual manner. Let

$\psi_1$ = bonding molecular orbital formed from $2p_x$ orbitals.

$\psi_2$ = bonding molecular orbital formed from $2p_z$ orbitals.

$\psi_3$ = antibonding molecular orbital formed from $2p_y$ orbitals.

Which of the following statements is/are correct?

**A)** The molecular orbital $\psi_1$ possesses a node containing the internuclear axis that originates from the angular node of the parent atomic orbitals.

**B)** The bonding molecular orbital $\psi_2$ contains a node passing through the internuclear axis.

**C)** The orbital $\psi_3$ contains a nodal plane perpendicular to the internuclear axis that arises due to destructive interference of atomic orbitals.

**D)** The number of nodes containing the internuclear axis is the same for bonding molecular orbitals formed from $2p_x$ and $2p_y$ orbitals.



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**27.** A non-stoichiometric iron oxide has the formula $Fe_{0.93}O_{1.00}$ and crystallizes in a cubic close-packed (ccp) array of oxide ions. The charge balance is maintained due to presence of $Fe^{3+}$ ions in the lattice. Which of the following statements is/are correct?

**A)** The percentage of total iron sites occupied by $Fe^{3+}$ ions is approximately 15.05%.

**B)** If the lattice constant remains unchanged, the ratio of density of the defective oxide to ideal $FeO$ is exactly 0.93.

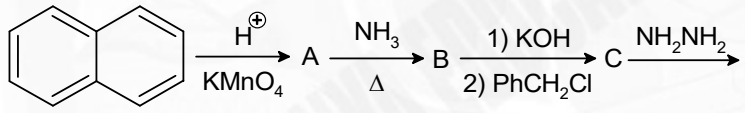
**C)** For every 100 oxide ions present in the crystal, the number of $Fe^{2+}$ ions is 79.

**D)** On dissolving this oxide in a non-oxidizing acid, the potential of a Pt electrode dipped in the solution is: $E = E^{\circ}\left(Fe^{3+}/Fe^{2+}\right) + 0.059\log(14/79)$ at 298 K relative to standard hydrogen electrode.

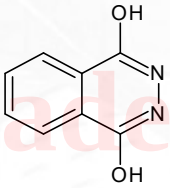
**28.** When a mixture of common salt and concentrated $H_2SO_4$ react with $MnO_2$, which of the following product(s) is/are formed?

**A)** $MnO$ **B)** $C\ell_2$ **C)** $NaHSO_4$ **D)** $O_2$

**29.**

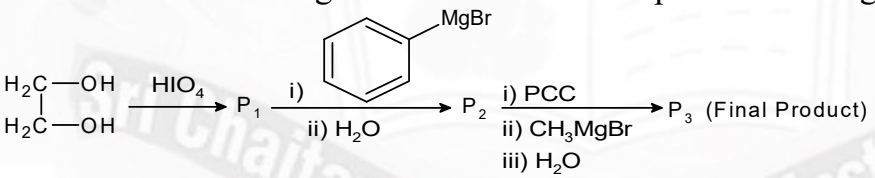



In the above reaction, product mixture contains mainly




**A)** $PhNH_2$ **B)** $PhNHNH_2$ **C)** **D)** $PhCH_2NH_2$

**30.** Which of the following is correct for the final product of the given sequence of reaction?




**A)** Compound $P_3$ on oxidation with $(CrO_3 + H_2SO_4)$ gives a compound which gives 2, 4–DNP test.

**B)** Compound $P_3$ on reaction with sodium hypoiodite gives yellow ppt.

**C)** Compound $P_3$ on reaction with cerric ammonium nitrate gives red colouration.

**D)** Compound $P_1$ on reaction with ammonial silver nitrate solution gives silver mirror



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## SECTION – III
## (NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. **(e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).**
**Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 3 (Max Marks: 18)**

- This Section contains **Six (06)** Questions. The answer to each question is a Numerical Value
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If ONLY the correct numerical value is entered)
  Zero Marks: 0 (In all other cases)

**31.** In a blast furnace, the reduction of hematite by carbon monoxide at **1000 K** is represented by the following simplified thermodynamic system:

1) $Fe_2O_3(s)+3CO(g)\rightleftharpoons 2Fe(s)+3CO_2(g)$ $\Delta G_1^\circ=-10\,kJ\,mol^{-1}$

2) $C(s)+CO_2(g)\rightleftharpoons 2CO(g)$ $\Delta G_2^\circ=+15\,kJ\,mol^{-1}$

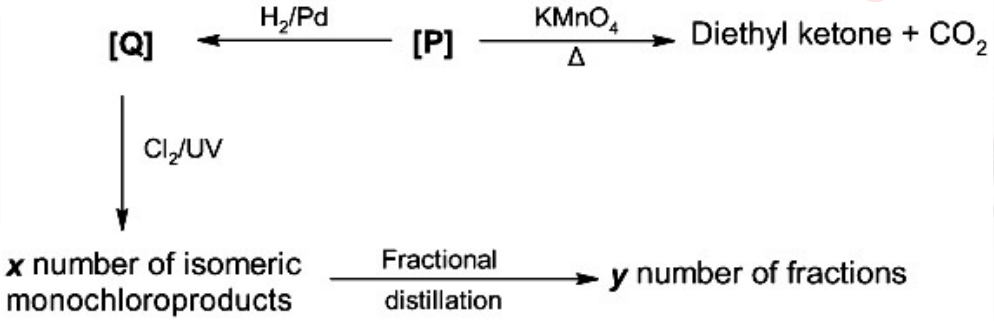
Magnitude of Gibbs free energy change $\left(\Delta G,\,in\,kJ\,mol^{-1}\right)$ for reaction (2) corresponding to the minimum ratio of partial pressures $P_{CO}/P_{CO_2}$ required in the furnace such that the reduction of $Fe_2O_3$ is just spontaneous at 1000 K, is **Z.** What is the value of Z/2?

Given: $P_{CO}+P_{CO_2}=2.49\,bar, R=8.3\,J\,mol^{-1}K^{-1}$

Use: $e^{10/24.9}=1.49$

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**32.**




If the major faction obtained on fractional distillation is subjected to further monobromination, what is the number of isomeric products (including stereo isomers) obtained?



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**33.** Benzene with 1-chlorododecane in the presence of anhydrous $A\ell C\ell_3$ forms **C**. Compound **C** with concentrated $H_2SO_4$ and subsequently neutralized with $NaOH$ to produce **D**. The surface tension of the solution of **D** in water is observed to stop decreasing significantly at a concentration of $1.2\times10^{-3}M$, indicating micelle formation. In a 1.0 L solution where the concentration of **D** is $5.2\times10^{-3}M$, the total number of micelles present is $3.011\times10^{19}$.
What is the value of $(P\times Q)/240$, where:

**P** = number of $sp^2$ hybridized carbon atoms in one molecule of surfactant **D**.
**Q** = aggregation number (number of surfactant molecules per micelle).
Take Avogadro constant: $N_A = 6.022\times10^{23}\,mol^{-1}$.
Assume all surfactant molecules above the critical micelle concentration are part of micelles.

**34.** Consider the neutral coordination complex tris(acetylacetonato)iron(III). In this complex.
Calculate the value of the integer **Z** defined as: $Z=\dfrac{\left\{\left(N\,rings+N_{sp^2}\right)-N_{m-C}\right\}}{3}$

Where: $N_{rings}$ = number of chelate rings formed in the complex
$N_{sp^2}$ = total number of $sp^2$ hybridized carbon atoms in the entire complex.
$N_{m-C}$ = total number of direct metal–carbon bonds in the complex.

**35.** A non-volatile compound **A** dissociates in solution to form soluble B, which is also non-volatile, according to: $A\rightarrow 2B$.
The dissociation follows first-order kinetics. Initially, 1 mol of A is dissolved in large excess of solvent. After 1 minute, the relative lowering of vapour pressure becomes 1.5 times its initial value. After how many minutes lowering of vapour pressure increases by 0.25 to that at one minute?

**36.** An empty, container of volume **V** is maintained at a constant temperature **T**. A pure liquid **A** is introduced into the container, establishing the heterogeneous equilibrium:
$A(\ell)\rightleftharpoons 2B(g)+C(g)$
At equilibrium, the total pressure in the container is 3 atm and contain 1.5moles of C. The volume of the container is then changed to 2V followed by addition of 3 mol of Argon while keeping the temperature constant. After some time, the heterogeneous equilibrium is reestablished. Let the new total pressure be $P_{total}$. Find the value of $(P_{total}-1)$.



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**MATHEMATICS** **Max Marks: 62**

## SECTION – I
## (SINGLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct. **Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases. Section 1 (Max Marks: 12)**

- Section 1 contains Four questions
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If the answer is correct)
  Zero Marks: **0** (If the question is unanswered)
  Negative Marks: -1 (In all other cases)

**37.** If a tangent of slope 4 of ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is normal to the circle $x^2+y^2+4x+1=0$ then the maximum value of $ab$ is

**A)** 4 **B)** 5 **C)** 6 **D)** 8

**38.** Area enclosed by the graph of the function $y=\ell n^2 x-1$ lying in the 4th quadrant, is

**A)** $\frac{2}{e}$ **B)** $\frac{4}{e}$ **C)** $2\left(e+\frac{1}{e}\right)$ **D)** $4\left(e-\frac{1}{e}\right)$

**39.** Let $A(2,-1,4)$ be a point in three–dimensional space. Let the plane P be given by $2x+y-z=7$ and the line L be defined as $\frac{x-1}{3}=\frac{y+2}{-2}=\frac{z-5}{1}$. The distance of the point A from the line L, measured parallel to the plane P, is equal to

**A)** $\frac{\sqrt{197}}{3}$ **B)** $\frac{\sqrt{251}}{4}$ **C)** $\frac{\sqrt{251}}{3}$ **D)** $\frac{7}{3}$

**40.** Let $f(x)$ be a differentiable function satisfying

$\sqrt[3]{f(x+y)}=\sqrt[3]{f(x)}+\sqrt[3]{f(y)}+1\ \forall x, y\in R$ and $f'(0)=3$, $h(x)=f(x)-x^3$ and

A = $\{\ell : \ell$ is a point where $h(|x|)$ is non–derivable$\}$.

Let $x_0$ is solution of the equation $f(x)=f^{-1}(x)$, then

$\frac{1}{\pi}\left(\cos^{-1}(\cos 2x_0)+4\tan^{-1}\left(\tan\frac{x_0}{2}\right)\right)+|A|$ is equal to ($|A|$ = Number of elements in A)

**A)** 0 **B)** 1 **C)** 3 **D)** 5



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## SECTION – II
## (ONE OR MORE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. **Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -1 in all wrong cases**

**Section 2 (Max Marks: 32)**

- Section two contains Eight (8) questions
- Each Question has Four Options and One Or More Than One of these four will be the correct answer.
- For each question, choose the option(s) corresponding to all the correct answer(s)

The Marking scheme to evaluate Answer to each question will be :
Full Marks: **+4** (If the answer is correct)
Partial Marks: **+3** (If all the four options are correct but **only 3** are chosen)
Partial Marks: **+2** (If THREE or More options are correct but **only 2** correct answers are chosen)
Partial Marks: **+1** (If TWO or More options are correct but **only 1** correct answer is chosen)
Zero Marks: **0** (If the question is unanswered)
Negative Marks: -1 (In all other cases)

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**41.** Let P be a square matrix of order 4 such that, $16P^4-96P^3+216P^2+81I_4=O$, and $\det(P)=16$ then which of following is/are **TRUE**? (Where $\det(P)$ represent determinant value of matrix $P,O$ is null matrix and $P^{-1}$ represents inverse of matrix $P$ and $adj(P)$ represents adjoint of matrix $P$)

**A)** $\det\left(adj\left(2I-3P^{-1}\right)\right)=3^6$ **B)** $adj(adjP)=2^8P$

**C)** $adj\left(adj(adjP)\right)=2^{28}P^{-1}$ **D)** $\det\left(adj\left(2I-3P^{-1}\right)\right)=\pm3^9$

**42.** Let $h(x)=\sin^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ and $g(x)=x^4-2x^3+x^2+4$.

$h(x)$ increases in $(-\infty,a_1)$; $h(x)$ decreases in $(a_1,\infty)$

$g(x)$ decreases in $(-\infty,a_1)$; $g(x)$ increases in $(a_1,a_2)$

$g(x)$ decreases in $(a_2,a_3)$; $g(x)$ increases in $(a_3,\infty)$

Let A and B be the point of interaction of tangent at $x=a_1$ to the curve $g(x)$ with the line $x=a_1$ and $x=a_3$ respectively.

Let D and C be the point of interaction of tangent at $x=a_2$ to the curve $g(x)$ with the line $x=a_1$ and $x=a_3$ respectively.

Join BC and AD with straight line. Which of the following is/are **TRUE**?



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**A)** Let number of points of non–differentiability of $y=|h(x)|$ be $a_4$. Then $2(a_1+a_2+a_3+a_4)=9$.

**B)** Area of quadrilateral ABCD is $\frac{1}{16}$

**C)** Area enclosed by $y=h(x)$ and x–axis is $\ell og_e 4$

**D)** Area enclosed by $y=h(x)$ and x–axis is $\frac{\pi}{2}-\ell n2$

**43.** Let $\alpha$ be a solution of the equation $2[x+32]=3[x-64]$ (where [x] is the greatest integer less than or equal to x) and let $\beta=\prod_{r=1}^{9}\sin\left(\frac{2r-1}{18}\right)\pi$, then which of the following can be **TRUE**?

**A)** $[\alpha]=[\beta]$ **B)** $\alpha=\frac{2051}{8}$ **C)** $[\alpha]\left[\frac{1}{\beta}\right]=1$ **D)** $\left[\frac{1}{\alpha}\right]+\left[\frac{1}{\beta}\right]=2^8$

**44.** Let $y=f(x)$ be a differentiable function passing through $M(1,-1)$ and has a property that the chord joining any two points $A(x_1,f(x_1))$ and $B(x_2,f(x_2))$ always intersects y–axis at $N(0,2x_1x_2)$. Then which of the following is/are **TRUE**?

**A)** $\int_0^{1/2} f(x)dx=\frac{1}{12}$

**B)** $\int_0^{1/2} f(x)dx=\frac{1}{24}$

**C)** Area bounded by $f(x)$ and x–axis is equal to $\frac{10}{3}$

**D)** Area bounded by $f(x)$ and x–axis is equal to $\frac{8}{3}$



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**45.** Let $A=\left[a_{ij}\right]$ be a square matrix of order 3 satisfying $A+adj(A)=\alpha A^T$. Identify which of the following statement(s) is (are) **CORRECT**?

**A)** If $\alpha=1$ then $\det(A)=0$

**B)** If $\alpha=1$ and $tr(A)=3$ then $tr(3A+adjA)=9$

**C)** If $\alpha=-2$ then $\det(3A+adjA)=0$

**D)** If $\alpha=-2$ and $tr(A)=3$ then $tr(4A+adjA)=6$

**46.** Let $<a_n>$ and $<b_n>$ be the arithmetic sequences each with common differences 2 such that $a_1<b_1$ and let $c_n=\sum_{k=1}^{n}a_k, d_n=\sum_{k=1}^{n}b_k$. Suppose the point $A_n(a_n,c_n)$ and $B_n(b_n,d_n)$ for every '$n$' $(n\geq 1)$ are lying on the parabola $C: y=px^2+qx+r$ where $p,q,r$ are constant. Then which of the following is/are **TRUE**?

**A)** $p=\frac{1}{4}$ **B)** $q=\frac{1}{2}$

**C)** If $r=0$, then $(a_1,b_1)=(0,2)$ **D)** If $r=0$, then $(a_1,b_1)=(1,3)$

**47.** Which of the following is (are) correct (where [.] represent GIF and {.} represent fractional part function)

**A)** Let $f(x)=\left\{\frac{x}{3}\right\}+\left[\frac{x}{5}\right], x\in(0,501)$ then number of points where $f(x)$ is discontinuous is 201.

**B)** Let $f(x)=[3\sin x+4\cos x], x\in[0,2\pi]$ then number of points where $f(x)$ is discontinuous is 18.

**C)** $\int_0^{\pi}|3\sin x+4\cos x|dx=10$

**D)** Let $f(x)=\frac{1}{x-1}+\frac{1}{x-2}+\frac{1}{x-3}+\frac{1}{x-4}$ then number of real roots of equation $f(x)=0$ is 3.



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**48.** Which of the following statement(s) is (are) **CORRECT**?

**A)** If $5\sec^{-1}\alpha+10\sin^{-1}\beta=10\pi$ then the value of $\tan^{-1}\alpha+\cos^{-1}(\beta-1)$ is $\frac{\pi}{4}$.

**B)** Minimum value of $\sin 2\pi x+\cos^2\pi x$ is $\frac{\sqrt{5}+1}{2}$

**C)** If $\sqrt{3}\tan 70^\circ=x+4\sin 70^\circ$ then $x$ is equal to 1.

**D)** If $f:(0,\pi)\to R$ be defined by $f(x)=\sum_{n=1}^{4}\left[1+\sin\frac{x}{n}\right]$ (where $[x]$ denotes largest integer less than or equal to $x$), then the sum of all the values in range of $f(x)$ is equal to 10.

## SECTION – III
## (NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. **(e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).**
**Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 3 (Max Marks: 18)**

- This Section contains **Six (06)** Questions. The answer to each question is a Numerical Value
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If ONLY the correct numerical value is entered)
  Zero Marks: 0 (In all other cases)

**49.** Let $A=\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{3}}+....+\frac{1}{\sqrt{10000}}$, then find $\left[\frac{[A]}{50}\right]$ _____ (where [.] denotes the greatest integer function)

**50.** If $n=\binom{30}{0}\binom{20}{10}+\binom{31}{1}\binom{19}{9}+\binom{32}{2}\binom{18}{8}+.....+\binom{40}{10}\binom{10}{0}$, then the number of trailing zeros at the end of $n$ is

**51.** Consider a equation $x^3-3x=\sqrt{x+2}$ such that $x$ is a real number then sum of its positive roots is $\lambda$ where $\left[\frac{3\lambda}{2}\right]$ is _____ ([.] denotes G.I.F.)



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**52.** If $a_1, a_2, a_3, \ldots\ldots$ are in HP. $a_1 > 0$ for each $i$ then $\sum_{i=1}^{n} \frac{n(a_i)^{\frac{2}{3}}(a_{i+1})^{\frac{2}{3}}}{(a_i)^{\frac{2}{3}} + (a_{i+1})^{\frac{2}{3}} + (a_i)^{\frac{1}{3}}(a_{i+1})^{\frac{1}{3}}}$ is $\frac{an^2}{b}$ where $a$ and $b$ are co–prime natural number then determine $2(b-a)$. (Where $a_1 = 1, a_{n+1} = 8$)

**53.** If $a, b, c \in N$, the probability that $\left(a^2 + b^2 + c^2\right)$ is divisible by 13 is $\frac{m}{n}$, where $m, n$ all relatively prime positive integers, then $(n-m)$ is

**54.** Let there be two circles with one touching x–axis at $A(2,0)$ and has radius 1 unit and other touching x–axis at $B(6,0)$ such that x–axis is the internal common tangent of two circles. Let C and D be the centre of first and second circles respectively. Again let $P(6,2)$ be a point on a line perpendicular to CD and bisecting AB. If PA and PB intersect circles at E and F respectively then find the length PF

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**4 Students in Top 11 in JEE-Advanced 2025, All India Open Category**

Rankers Academy JEE

| | | | | | |
|---|---|---|---|---|---|
| 16 RANK<br>DEVDUTTA MAJHI<br>HT. No. 255053116* | 18 RANK<br>DHARMANA GNANA RUTVIK SAI<br>HT. No. 256055278 | 19 RANK<br>VANGALA AJAY REDDY<br>HT. No. 256131009 | 23 RANK<br>AKSH GOGI<br>HT. No. 252071075* | 26 RANK<br>P HEMA SAI SURYA KARTHIK<br>HT. No. 256033006 | 27 RANK<br>SARKARSAMUDRA<br>HT. No. 252071105* |
| 30 RANK<br>OM PRAKASH BEHERA<br>HT. No. 252021018* | 32 RANK<br>SUNKARA SAI RISHANTH REDDY<br>HT. No. 256165327 | 34 RANK<br>DHRUBA JYOTHI PANJA<br>HT. No. 252048248* | 35 RANK<br>BHAVESH JAYANTHI<br>HT. No. 251043080 | 36 RANK<br>ADVAY MAYANK<br>HT. No. 252104113* | 37 RANK<br>KARMANYA GUPTA<br>HT. No. 252081477* |
| 42 RANK<br>MD ANAS<br>HT. No. 252046210* | 45 RANK<br>RAMIT GOYAL<br>HT. No. 257001113* | 52 RANK<br>MAULIK JAIN<br>HT. No. 252079407* | 54 RANK<br>GARV<br>HT. No. 252056188* | 59 RANK<br>LARISSA<br>HT. No. 252079071* | 60 RANK<br>ARYAN BALABADRULA<br>HT. No. 256132077 |
| 63 RANK<br>SAMYAJYOTI BISWAS<br>HT. No. 255058456* | 64 RANK<br>AARUSH ANAND<br>HT. No. 251006176* | 72 RANK<br>RUSHMITH BANDARI<br>HT. No. 256168048 | 78 RANK<br>KORIKANA RASAGNYA<br>HT. No. 256057046 | 87 RANK<br>LAKSHYA SHARMA<br>HT. No. 252070079* | 91 RANK<br>AVANEESH BANSAL<br>HT. No. 251113130* | 95 RANK<br>KAVYA AGGARWAL<br>HT. No. 252079121* |

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|---|---|---|---|---|
| BELOW 100 ALL INDIA OPEN CATEGORY RANKS | 29 | BELOW 500 ALL INDIA OPEN CATEGORY RANKS | 113 | BELOW 1000 ALL INDIA OPEN CATEGORY RANKS | 205 | BELOW 1000 ALL INDIA CATEGORY RANKS COUNT | 745 | NUMBER OF QUALIFIED RANKS | 4,212 |



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